

KuanYen(Rob) Huang

Pittsburgh PA | kuanyenh@andrew.cmu.edu | +1(412)475-7522 | www.linkedin.com/in/Rob-huang/ | 12oi3.github.io

PROFESSIONAL SUMMARY

Graduate student at Carnegie Mellon University with 8 years of game development experience specializing in technical game design, gameplay systems, and Unity tool development. Experienced in designing modular gameplay architectures, scalable UI systems, and rapid iteration workflows driven by playtesting and cross-disciplinary collaboration.

Education

Carnegie Mellon University
Master of Entertainment Technology

Pittsburgh, PA
Aug 2025 - May 2027

National Tsing Hua University
Bachelor of Science in Computer Science

Hsinchu, TW
Sep 2020 - Jun 2024

Experience

CreSpirit - A startup indie game company specializing in Metroidvania titles.

Hsinchu, TW

UI/UX Tool Programmer

Nov 2024 - Jun 2025

- Developed the contracted Unity plugin “CreSpiritTalker” for dialogue and visual novel interfaces.
- Built a customizable Unity editor tool with editor windows for dialogue databases and workflow management.
- Engineered an embedded text command system supporting developer-defined dialogue behaviors.
- Implemented a save migration framework to maintain compatibility across newer system versions.

Technical Game Designer Intern

Feb 2024 – Jun 2024

- Engineered and designed a mini-game for Last Command: B-Side with a built-in level editor to support future iteration.
- Applied the MVC architecture pattern to accelerate gameplay iteration from playtesting feedback.
- Contributed to narrative structure planning and multilingual database management.

Projects

Project Fusion - Auto battler prototype featuring artifact fusion mechanics within a roguelike framework

Technical Game Designer

Apr 2023 - May 2024

- Led project development by architecting modular gameplay systems and designing mechanics around core experience.
- Recreated Unreal Engine’s Gameplay Ability System architecture in Unity for scalable combat systems.
- Designed a policy-based passive ability framework to support modular combat behaviors.
- Implemented an MVC-based UI architecture to reduce coupling and improve maintainability.

You Ken-Do It - VR kendo game focused on block-and-counter combat mechanics

Technical Game Designer

Mar 2023 - Jun 2023

- Directed development of a VR kendo experience for Meta Quest 2 in Unity for gaming and training purposes.
- Improved VR UI/UX through iterative playtest-driven adjustments for readability and immersion.
- Designed core combat mechanics like slicing, parrying, and counterattack for responsive combat interactions.

Survivor For the Feast - Survivor-like game featuring hunting, gathering, and cooking systems

Technical Game Designer

Sep 2022 - Jan 2023

- Led a 5-member team using agile workflows and Discord-based production management.
- Designed core gameplay loops and gameplay systems including combat skills, progression mechanics, and balancing.
- Coordinated milestone planning and cross-functional collaboration within a 4-month timeline.
- Won Best Game Design at the Taiwan Student Innovation Game Design Competition among 26 teams.

Skills

Languages & Tools: C# | C++ | Python | C | Unity | VR | Git | Perforce | Godot | Notion | Spreadsheet Tools

Specializations: Technical Game Design | UI/UX | Tool Development | Agile Workflow | Playtesting | Design Patterns